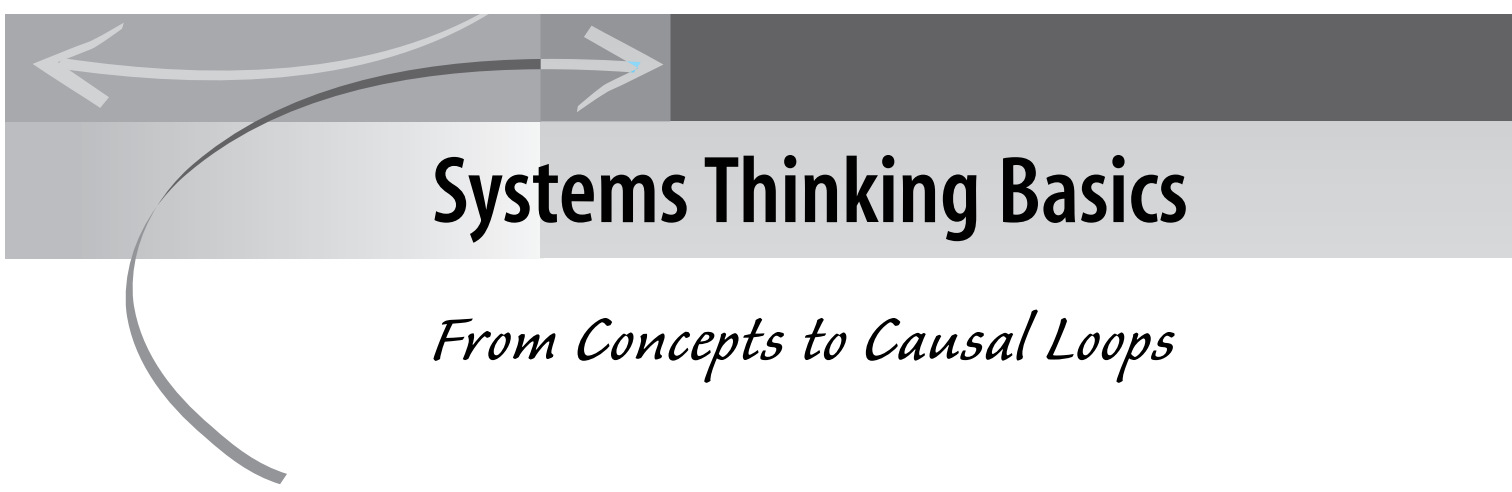


# **Systems Thinking Basics**



A decorative header consisting of a horizontal bar divided into three sections of different shades of gray. A white arrow points left from the left section to the middle section, and another white arrow points right from the middle section to the right section. A large, thin, curved gray line starts from the left side of the bar, loops around the left side of the title, and ends near the bottom left of the page.

# Systems Thinking Basics

*From Concepts to Causal Loops*

Virginia Anderson  
AND Lauren Johnson



PEGASUS COMMUNICATIONS, INC.  
Westford, Massachusetts

© 1997 by Pegasus Communications, Inc.  
First printing March 1997

All rights reserved. No part of this publication may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopying and recording, or by any information storage or retrieval system, without written permission from the publisher.  
For additional copies, information about other titles published by Pegasus Communications, Inc., or general inquiries, contact:

**PEGASUS COMMUNICATIONS, INC.**

319 Littleton Road, Suite 103

Westford, MA 01886 USA

Phone 800-272-0945 / 781-398-9700

Fax 781-894-7175

customerservice@pegasuscom.com    info@pegasuscom.com

**www.pegasuscom.com**

Library of Congress Cataloging-in-Publication Data

Anderson, Virginia, 1947–

Systems Thinking Basics: From Concepts to Causal Loops / by Virginia Anderson and Lauren Johnson.

p.    cm.

ISBN 1-883823-12-9

1. Industrial management—Case studies—Problems, exercises, etc.

2. Systems analysis—Problems, exercises, etc.    3. Decision making—exercises, etc.

I. Johnson, Lauren.    II. Title.

HD30. 19. A53    1997

658.4'032—dc21

96-39762

CIP

Acquiring editor: Kellie Wardman O'Reilly

Project editor: Lauren Johnson

Production, art, and design: Boynton Hue Studio

First edition.

To the Reader: Why Use This Book?   vii

## **SECTION 1   WHAT ARE SYSTEMS?   1**

What Is a System?   2

Events, Patterns, Structure   5

**LEARNING ACTIVITIES   10**

1: Identifying Systems   10

2: Reflecting on System Characteristics   13

3: Understanding System Feedback   14

## **SECTION 2   WHAT IS SYSTEMS THINKING?   17**

The Principles of Systems Thinking   18

Systems Thinking as a Special Language   20

**LEARNING ACTIVITIES   22**

1: Stretching the Timeline   22

2: The Shape of the Problem   25

3: Is Time on Your Side?   29

4: From Short Term to Long Term   30

5: In the Midst of a Problem   34

## **SECTION 3   UNCOVERING SYSTEMIC STRUCTURES: DRAWING BEHAVIOR OVER TIME GRAPHS   37**

Formulating a Problem   38

Identifying Variables   40

Drawing Behavior Over Time Graphs   42

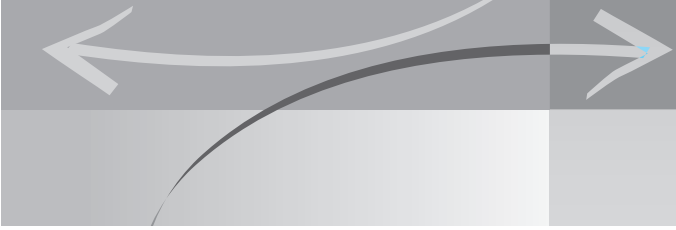
**LEARNING ACTIVITIES   45**

1: The Problem with Price Promotions   45

2: The Case of the Energy Drain   46

3: The Case of the Audio-Electronic Roller Coaster   48

|                    |                                                                                    |            |
|--------------------|------------------------------------------------------------------------------------|------------|
| <b>SECTION 4</b>   | <b><i>UNCOVERING SYSTEMIC STRUCTURES:<br/>BUILDING CAUSAL LOOP DIAGRAMS</i></b>    | <b>51</b>  |
|                    | Anatomy of a Causal Loop Diagram                                                   | 52         |
|                    | Building a Causal Loop Diagram                                                     | 58         |
|                    | About Multiloop Diagrams                                                           | 64         |
|                    | <b>LEARNING ACTIVITIES</b>                                                         | 65         |
|                    | 1: The Case of the Plateauing Profits                                              | 65         |
|                    | 2: The Case of the Collapsing Banks                                                | 66         |
|                    | 3: The “Organic To Go” Story                                                       | 67         |
|                    | 4: The Case of the Restricted Revenues                                             | 70         |
|                    | 5: The All-for-One Cooperative                                                     | 73         |
|                    | 6: The Problem with Used CDs                                                       | 75         |
| <b>SECTION 5</b>   | <b><i>COMPLEX SYSTEMS</i></b>                                                      | <b>77</b>  |
|                    | Characteristics of Complex Systems                                                 | 78         |
|                    | The Strengths and Weaknesses of Complex Systems                                    | 79         |
|                    | ComputeFast: The Story of a Complex System                                         | 80         |
| <b>SECTION 6</b>   | <b><i>FROM LOOPS TO LEVERAGE:<br/>APPLYING AND PRACTICING SYSTEMS THINKING</i></b> | <b>87</b>  |
|                    | The Do’s and Don’ts of Systems Thinking on the Job                                 | 88         |
|                    | Practicing Life-Long Systems Thinking                                              | 91         |
|                    | The Learning Journey                                                               | 94         |
| <b>APPENDIX A:</b> | <b><i>ADDITIONAL LEARNING ACTIVITIES</i></b>                                       | <b>95</b>  |
| <b>APPENDIX B:</b> | <b><i>LEARNING ACTIVITY KEY POINTS AND SUGGESTED RESPONSES</i></b>                 | <b>105</b> |
| <b>APPENDIX C:</b> | <b><i>A PALETTE OF SYSTEMS THINKING TOOLS</i></b>                                  | <b>119</b> |
| <b>APPENDIX D:</b> | <b><i>THE SYSTEMS ARCHETYPES</i></b>                                               | <b>123</b> |
| <b>APPENDIX E:</b> | <b><i>A GLOSSARY OF SYSTEMS THINKING TERMS</i></b>                                 | <b>127</b> |
| <b>APPENDIX F:</b> | <b><i>ADDITIONAL RESOURCES</i></b>                                                 | <b>131</b> |



## TO THE READER

### Why Use This Book?

Congratulations! By picking up this book, you've taken the first step in learning to use the powerful array of systems thinking tools. These tools offer a fresh, highly effective way to grasp the complexities of organizational life and to address the stubborn problems that often confront us in the business world. *Systems Thinking Basics: From Concepts to Causal Loops* is designed to help you discover the principles of systems and systems thinking and begin actually using systems thinking tools. There is a broad palette of systems thinking tools available; this book emphasizes two foundational devices: *behavior over time graphs* and *causal loop diagrams*.

#### **Why Systems Thinking?**

Why take time out of your busy schedule to read this book? Because systems thinking offers a valuable new perspective on our most persistent organizational problems and our role in them. One of the major points that systems thinking makes is that everything—and everyone—is interconnected in an infinitely complex network of systems. When we begin to see the world—and one another—through this “lens,” we start seeing our circumstances in a new light, taking more responsibility for our own role in problems, and identifying more effective ways of addressing recurring difficulties. Systems thinking has a power and a potential that, once you've been introduced, are hard to resist. You'll learn more about the benefits of systems thinking in Section 2.

#### **How to Use This Book**

The world of systems and systems thinking marks a dramatic shift from the more linear, analytic way of thinking that many of us are used to. It's especially challenging to convey the abstract characteristics of systems in print. Because systems themselves are so complex and so interconnected with everything around them, it's also impossible to capture any one entire system on paper. For this reason, the tools and activities in this book are meant to offer primarily a glimpse into systems behavior.

We also hope you will see the book as the starting point to a long, learning-filled journey. Systems research and theories are constantly evolving, and the concepts and tools in this book offer only one of many methods for looking at the whole. When you finish the book, you can continue your adventure into exploring systems thinking tools by using some of the resources listed in Appendix F.

In writing *Systems Thinking Basics*, we not only needed to decide what tools to include; we had to choose a plan for sequencing the sections. We

opted for a format that allows each section to build on the one that came before, because that arrangement seemed best suited for new systems thinkers. For this reason, we recommend that you read the sections in numerical order, although you are certainly free to choose a different order depending on your interests and your familiarity with the material. We also hope that you'll work as often as possible with others on the readings and activities—it's when people use systems thinking tools *together* that these tools are their most potent.

Your journey through *Systems Thinking Basics* will begin with an exploration of the unique characteristics of systems (Section 1), and then move on to a definition of systems thinking and an explanation of its importance, especially to organizational life (Section 2). In sections 3 and 4, you will have the opportunity to create your own diagrams of systemic structures, step by step. Section 5 then gives you a taste of complex systems. Finally, Section 6 answers the question, "What next?," by offering advice on practicing systems thinking in your day-to-day life.

### ***About the Learning Activities***

Mastering systems thinking tools requires practice with lots of real-life examples. Accordingly, each section in this book contains a wealth of illustrations from the business world. Most sections then present a series of learning activities that challenge you to apply your new knowledge. The learning activities can be done as self-study or in groups, but again, we encourage you to work in groups as much as possible. We invite you to share with each other your responses to the activities and the insights you gain as you work with the activities.

The learning activities range from reflection and discussion to graphing and diagramming, and are filled with case studies adapted from recent analyses published in leading business journals, including *The Systems Thinker*<sup>TM</sup> newsletter. There is also a section (Appendix A) that offers extra learning activities should you desire additional practice. For the majority of the learning activities, you can check your responses against the "Learning Activity Key Points and Suggested Responses" in Appendix B. Remember, however, that there is no one right way to describe and diagram a system. These activities, and the suggested responses, are meant to spark your imagination and serve as a starting point for you to think about systems.

### ***Acknowledgments***

The work contained in this book is built on the prior work of many others. First, we acknowledge Peter Senge for introducing the concept of systems thinking to a broad management audience in *The Fifth Discipline*. In many ways, the tremendous success of his book is what made the writing of this workbook possible in the first place. We also acknowledge the many people who were involved in the development of systems thinking principles, concepts, and tools before they ever appeared in *The Fifth Discipline*, including John Sterman, Dennis Meadows, Dana Meadows, Michael Goodman, David Kreutzer, Jenny Kemeny, Jack Pugh, George Richardson, James Lyneis,

Nathaniel Mass, Barry Richmond, and others. In addition, we thank all the systems thinking workshop participants and contributors to *The Systems Thinker* newsletter, who have given us valuable insights into understanding and applying systems thinking to organizational life. Finally, we honor Jay Forrester's brilliance and insight in founding the field of system dynamics, which we believe has much to contribute to the practice of management.

We'd like to thank the following colleagues who kindly reviewed the manuscript for this book. Each of them offered valuable insights and thoughtful suggestions that greatly strengthened the book:

Turina Bakken, MAQIN

Polly Bashore, General Motors

Dave Buffenbarger, New Dimensions in Learning

Lou Ann Daly, Innovation Associates

Richard Karash, Speaker, Facilitator, Trainer, "Towards Learning Organizations"

Daniel H. Kim, The MIT Center for Organizational Learning, and Pegasus Communications

Keith Perry, McClellan Air Force Base

Kellie Wardman O'Reilly, Pegasus Communications

Carol Ann Zulauf, Suffolk University

and, finally, all the folks at Coca-Cola:

Michael Canning, Connie Carroad, Dianne Culhane, Teresa Hogan, Scott Moyer, Cheryl Oates, and Rodolfo Salgado

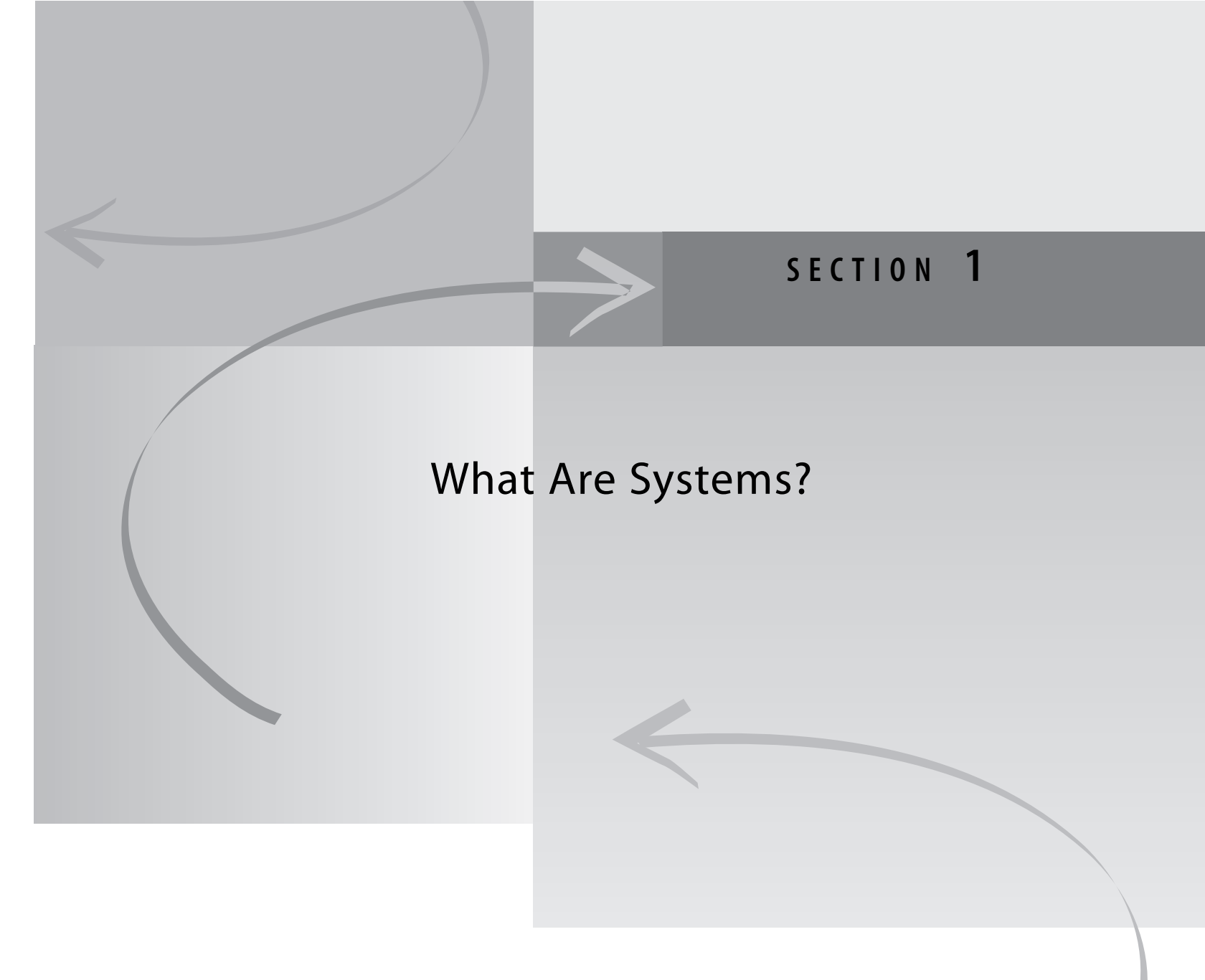
**W**e hope you'll find your adventure into systems thinking stimulating and rewarding on both a personal and a professional level. We also invite you to contact us with any comments, questions, or suggestions about using this book or about systems thinking in general.

Bon voyage!

Virginia ("Prinny") Anderson (VRAnderson@aol.com)

Lauren Johnson (scott-laurie@att.net)



The background of the top half of the page features a light gray grid. Overlaid on this grid are several large, thin, curved arrows in a darker gray shade. One arrow starts in the top-left quadrant and points towards the top-right. Another starts in the top-left and curves around to point towards the center. A third starts in the bottom-left and curves around to point towards the center. A fourth starts in the bottom-right and points towards the center. The text 'SECTION 1' is positioned in the top-right quadrant, within a darker gray horizontal band.

## SECTION 1

# What Are Systems?

**W**elcome to the world of systems and systems thinking! You may be asking yourself, Why is it important to explore systems? One reason is that we live in and are influenced by systems all around us, from the natural environment to healthcare, education, government, and family and organizational life. Understanding how these systems work lets us function more effectively and proactively within them. The more we build our understanding of system behavior, the more we can anticipate that behavior and work *with* the system to shape the quality of our lives.

This section introduces you to the idea of systems and what makes them unique. In the learning activities at the end of the section, you will have the opportunity to identify some major systems in your own work life and to think about typical system behavior.



## WHAT IS A SYSTEM?

A system is a group of interacting, interrelated, or interdependent components that form a complex and unified whole. A system's components can be physical objects that you can touch, such as the various parts that make up a car. The components can also be *intangible*, such as processes; relationships; company policies; information flows; interpersonal interactions; and internal states of mind such as feelings, values, and beliefs.

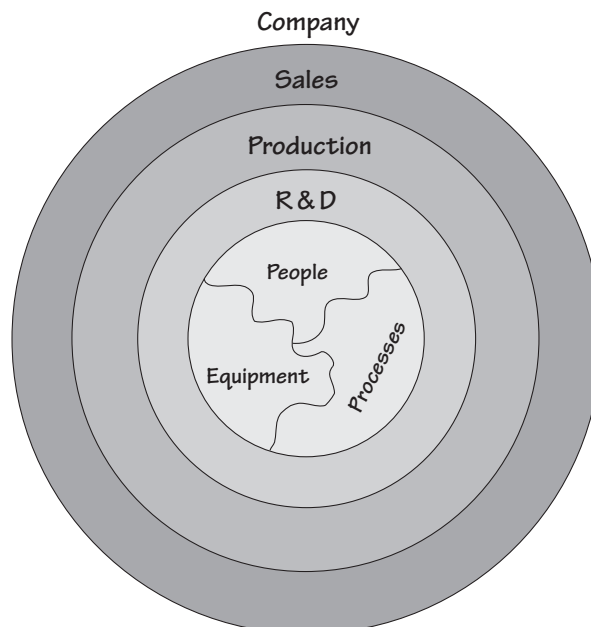
In an organizational setting, for example, the R&D group is a system made up of people, equipment, and processes that create new products to be manufactured by the production system and sold by the sales system. The components of the R&D group have to interact with one another to perform their function and thus are interdependent. In turn, the R&D group interacts and is interdependent with other systems within the company. A system such as the R&D group always has a specific purpose in relation to an even larger system—in this case, the entire organization (Figure 1.1, “Interdependent Systems Within Interdependent Systems”).

Your body is another example. Within it, your circulatory system delivers oxygen, nutrients, hormones, and antibodies produced by other systems and carries waste to the excretory system. The circulatory system is made up of the heart, veins and arteries, blood, and a host of supporting elements. All of these components interact to carry out their purpose within the larger system—your entire body.

Both of these examples raise an intriguing point about systems: We can think of all systems as nodes embedded in a giant network in which everything is connected. For example, the company described above, with its interdependent R&D, production, and sales systems, is itself a large system that is interdependent with an even larger system—industry as a whole.

FIGURE 1.1

Interdependent Systems Within Interdependent Systems



And industry is interdependent with an even larger system—the economy—and so on. The more we widen our view in this way, the more we see that everything—from the tiniest subatomic particle to the universe (and maybe beyond!)—is intertwined.

We can also distinguish between natural systems and human-made, nonliving systems. Natural systems—a living being's body, human societies, an ecosystem such as a prairie—have an enormous number and complexity of components and interactions among those components. They also have virtually an infinite number of connections to all the systems around them. Human-made systems—cars, for example—can also be quite complex, but these nonliving systems are not as intricately linked to systems around them. If a car breaks down, the impact of this event is not nearly as far-reaching as if a species were removed from a prairie ecosystem (although you may disagree if it's *your* car that breaks down!). Put another way, human-made nonliving systems are more self-contained than natural systems, which we can think of as more open in their connections to surrounding systems.

## ***Defining Characteristics of Systems***

Systems have several essential characteristics:

### **1. A system's parts must all be present for the system to carry out its purpose optimally.**

If you can take components away from something without affecting its functioning and its relationships, then you have just a collection, not a system. For example, if you remove a cashew from a bowl of mixed nuts, you have fewer nuts, but you have not changed the nature of the collection of components. Therefore, a bowl of mixed nuts is not a system.

Similarly, if you can add components to a collection without affecting its functioning and relationships, it's still just a collection. So, if you add pistachios to your bowl of mixed nuts, you have more nuts and you have a different mix, but you still have just a collection of nuts.

However, if you assign new tasks to an R&D group or redefine the job descriptions of its staff, you will likely change the group's functioning and relationships—whether for the better or worse. The R&D group is not just an assortment of people, equipment, and processes; it is a system.

### **2. A system's parts must be arranged in a specific way for the system to carry out its purpose.**

If the components of a collection can be combined in any random order, then they do not make up a system. For example, in a bowl of fruit, the oranges can go at the bottom, in the middle, or on the top without changing the essential nature of the collection of fruit.

However, in a system such as a company, imagine what would happen if the parts shifted around randomly—if, for instance, the accounting specialists suddenly decided to work on the production line, and the production specialists decided to write marketing copy. Of course, people do change jobs within their companies, but only after training and much transition time. Most companies function best when people are working in jobs

that match their skills and experience, and when the staff is organized according to a specific plan.

### **3. Systems have specific purposes within larger systems.**

All systems have a specific purpose in relationship to the larger system in which they're embedded, as we saw in the examples of the R&D department and human circulatory system above. Because each system has its own purpose, each is a discrete entity and has a kind of integrity that holds it together. In other words, you can't force two or more systems together and get a new, single, larger system. Nor can you subdivide a system and automatically end up with two smaller identical, functioning systems.

As the saying goes, if you divide an elephant in half, you don't end up with two smaller elephants. And if you put two small elephants together, you don't have a new, single, larger elephant (although some day you may end up with a new system—known as a herd!).

### **4. Systems maintain their stability through fluctuations and adjustments.**

Left to themselves, systems seek to maintain their stability. Your organization does its best to maintain a designated profit margin just as most human bodies work to maintain a temperature of about 98.6 degrees Fahrenheit. If you examined your organization's revenues against expenditures every week or graphed your body temperature every five minutes, you would probably draw a wobbly, fluctuating line that nevertheless holds steady *overall*. Margins appear and disappear as a company pays its suppliers and collects checks from customers. Your body temperature rises and falls depending on your mood and your level of physical exertion. On average, however, your body temperature remains stable. And, with reasonable management and no cataclysmic change, your organization's margin also remains stable overall.

Systems achieve this stability through the interactions, feedback, and adjustments that continually circulate among the system parts, and between the system and its environment. Let's say a corporation receives an unusually large stack of suppliers' invoices (external stimuli) in the mail. The accounts payable department responds by paying the bills. As the checks go out the door, the accounting department, alarmed, compares revenue versus expenditures and gives feedback to management: Expenditures are up and revenues aren't covering them. Management then adjusts the system by reminding key customers to pay overdue invoices. Similarly, if you go for a run, your exertion warms your body. The sensation of heat is fed back to your sweat glands, which begin to work. Over time, sweating readjusts your temperature back to the norm.

### **5. Systems have feedback.**

Feedback is the transmission and return of information. For example, imagine that you are steering your car into a curve. If you turn too sharply, you receive visual cues and internal sensations that inform you that you are turning too much. You then make adjustments to correct the degree of your turn. The most important feature of feedback is that it provides the catalyst for a change in behavior.

A system has feedback within itself. But because all systems are part of larger systems, a system also has feedback between itself and external systems. In some systems, the feedback and adjustment processes happen so quickly that it is relatively easy for an observer to follow. In other systems, it may take a long time before the feedback is returned, so an observer would have trouble identifying the action that prompted the feedback. For example, if you sunbathed a lot in your teens, you may develop skin problems after age 40—but because so much time passed between the two events, you may not recognize the connection between them.

Finally, feedback is not necessarily transmitted and returned through the same system component—or even through the same system. It may travel through several intervening components within the system first, or return from an external system, before finally arriving again at the component where it started.

For instance, imagine that the company you work for is suffering financially and decides to lay off 20 percent of the work force. That quarter, the layoff does indeed improve the looks of the financial bottom line. On this basis, the upper management might decide that layoffs are a reliable way to improve the financial picture.

However, let's say you survived the layoff; how would you describe your state of mind and that of your other remaining colleagues? Besides cutting costs, layoffs are also famous for damaging morale and driving people to “jump ship” in search of more secure waters. Eventually, as low morale persists, you and your colleagues might start coming to work late and leaving early, and caring less and less about the quality of your work. Productivity could drop. In addition, everyone who leaves—whether voluntarily or by being laid off—takes valuable skills and experience with them, so the overall capability of the work force goes down, further hurting productivity. Lowered productivity leads to expensive mistakes and lost sales from disgruntled customers. All this eats away even more at the company's revenue, tempting management to think about having even more layoffs to cut costs.

In this example, the feedback that made layoffs look like good policy was returned quickly—probably within one quarter. The feedback about the long-term costs of layoffs went through more steps and took a lot longer to return. Yet this information was essential for the management team to see the full impact of their decisions.



## ***EVENTS, PATTERNS, STRUCTURE***

In reading all this information, you may be wondering what actually gives rise to systems. Systems are built on structures that leave evidence of their presence, like fingerprints or tire marks, even if you can't see them. But what is structure, exactly? The concept is difficult to describe. In simplest terms, structure is the overall way in which the system components are interrelated—the organization of a system. Because structure is defined by the *interrelationships* of a system's parts, and not the parts themselves, structure is invisible. (As we'll see later, however, there are ways to draw our *understanding* of a system's structure.)

Why is it important to understand a system's structure? Because it's system structure that gives rise to—that explains—all the events and trends that we see happening in the world around us.

Perhaps the best way to grasp the role of structure is to explore the Events / Patterns / Structure pyramid, shown in Figure 1.2.

## Events

We live in an event-focused society (Figure 1.3, “The Tip of the Pyramid”). A fire breaks out in the neighborhood; a project misses a deadline; a machine breaks down. We tend to focus on events rather than think about their causes or how they fit into a larger pattern. This isn't surprising; in our evolutionary development as a species, this ability to respond to immediate events ensured our very survival.

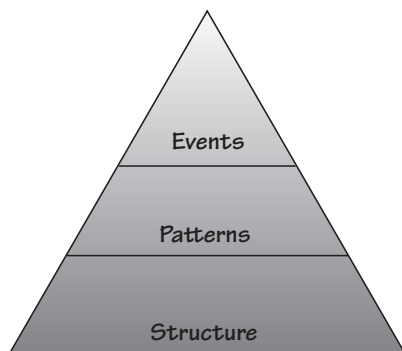
But focusing on events is like wearing blinders: You can only *react* to each new event rather than anticipate and shape them. What's more, solutions designed at the event level tend to be short lived. Most important, they do nothing to alter the fundamental structure that caused that event. For example, if a building is burning, you would want local firefighters to react by putting out the fire. This is a necessary and essential action. However, if it is the *only* action ever taken, it is inadequate from a systems thinking perspective. Why? Because it has solved the immediate problem but hasn't changed the underlying structure that caused the fire, such as inadequate building codes, lack of sprinkler systems, and so on.

By uncovering the elusive systemic structure that drives events, you can begin identifying higher-leverage actions. The next step to comprehending systemic structure is to move from thinking at the event level to thinking at the pattern level.

## Patterns

Whereas events are like a snapshot, a picture of a single moment in time, patterns let us understand reality at a deeper level (Figure 1.4, “Moving from Events to Patterns”). Patterns are trends, or changes in events over

**FIGURE 1.2**  
The Events / Patterns / Structure Pyramid



**FIGURE 1.3**  
The Tip of the Pyramid



**FIGURE 1.4**  
Moving from Events to Patterns

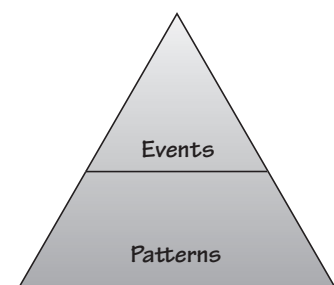
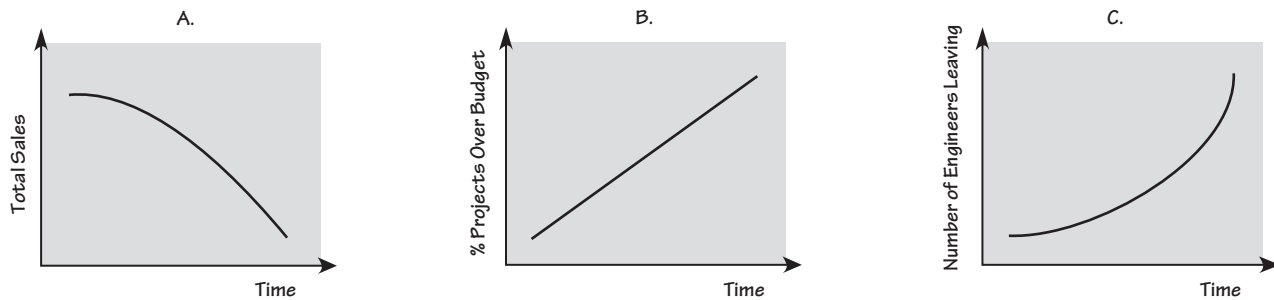


FIGURE 1.5

## Graphs of Patterns



time. Whenever you see a pattern of events—for example, sales have been declining over the past few years, or two-thirds of the department’s projects have gone over budget in the last year, or several senior engineers have left the company recently, most of them in the last six months—you’re getting one step closer to grasping the systemic structure driving that pattern.

In each of the above examples, you could draw a simple graph to represent the trend (Figure 1.5, “Graphs of Patterns”).

What is the advantage of thinking at the pattern level, as opposed to the event level? Detecting a pattern helps you put the most recent event in the context of other, similar events. The spotlight is then taken off the specific event, and you can focus on exploring how the series of events are related and begin thinking about what caused them. In the end, to anticipate events and ultimately change a pattern, you need to shift your thinking one more time: to the level of structure (Figure 1.6, “The Complete Pyramid”).

## Structure

To move to this deeper level of understanding, let’s reconsider the above example of the senior engineers’ exodus. You might begin digging for the structure behind this pattern by asking, “What’s causing more and more senior engineers to leave?” In this case, suppose a change in corporate policy has cut both the budget and the number of administrative assistants for the engineering group. The engineers’ workloads have ballooned, and they’ve begun grumbling more and more about their job pressure. Worse yet, as some of them leave, those left behind get even more upset as their workloads expand further. It’s a vicious cycle that you might sketch as shown in Figure 1.7, “The Engineering Exodus,” p. 8.

FIGURE 1.6

## The Complete Pyramid

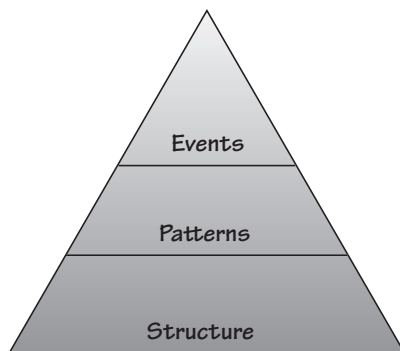
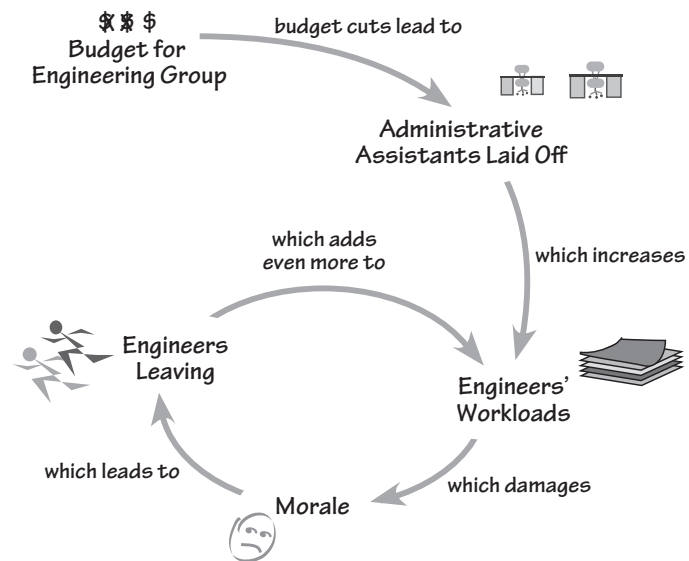


FIGURE 1.7

## The Engineering Exodus



Whenever we ask questions like, “Why is this pattern happening?” or “What’s causing these events?” we are probing at structure. Thinking at the structural level means thinking in terms of causal connections. *It is the structural level that holds the key to lasting, high-leverage change.* Let’s return to our example about a house catching fire, to see how this works. To fight fires at the event level, you would simply *react* to quell the fire as soon as possible after it broke out. You would probably then repair any smoke and water damage, and put the incident out of your mind.

## A NOTE ABOUT DIAGRAMMING SYSTEMS

As we saw earlier, all systems are part of larger systems, so it’s impossible to capture any system in its entirety on paper. Nevertheless, there are ways of depicting *parts* of systems in a diagram, in order to glimpse how a system works and how you might alter its behavior. One such way is to create a causal loop diagram, or CLD. (Figure 1.7 is an example of this kind of drawing.) These diagrams provide a starting place for discussing and thinking about problematic events or patterns, and for opening the door to addressing problems differently. In particular, they help you gain insight into systemic structures, and they identify ways you might change the system’s behavior. After all, it is changes made at the system level, rather than at the pattern or event level, that often prove to be the most long-lasting and self-sustaining.

It’s important to remember, however, that graphic representations of systems are just that: representa-

tions. As you will see later in this book, there is no one right way to draw a causal loop diagram or even to describe an entire system. Any diagram that you draw reflects your own assumptions about the system, and is limited to what you define as the most pertinent part of the system you’re studying. This is why working in groups is so beneficial—you gain insights from the multiple perspectives.

Later in this book, you’ll have the opportunity to practice drawing CLDs. As with all the activities in this book, we encourage you to work as a group whenever possible in creating CLDs. A causal loop diagram generated by a group is especially valuable because it reveals the interplay of each group member’s perspective on the system in question. The process of constructing the drawing encourages group members to share their assumptions and understandings about the issue at hand. The more this kind of sharing happens, the more insights get sparked.

How would you fight fires at the pattern level? You would begin anticipating *where* other fires are most likely to occur. You may notice that certain neighborhoods seem to suffer more fires than others. You might locate more fire stations in those areas, and staff them based on past patterns of usage. By doing these things, you would be able to fight fires more effectively by *adapting* to the *patterns* you have observed.

However, your actions haven't done anything to reduce the actual occurrence of fires. To address the problem at this level, you need to think about the structure that gives rise to the pattern of fires. At the systemic structure level, you would ask questions like, "Are smoke detectors being used? What kinds of building materials are least flammable? What safety features reduce fatalities?" Actions that you take at this deep level can actually cut down the number and severity of fires. Establishing fire codes with requirements such as automatic sprinkler systems, fire-proof materials, fire walls, and fire alarm systems saves lives by preventing or containing fires.

Here's where the real power of structural-level thinking comes in: Actions taken at this level are *creative*, because they help you to shape a different future, the future that you want. Does this mean that high-leverage actions can be found only at the structural level? No—leverage is a relative concept, not an absolute. Our ability to influence the *future* increases as we move from event-level to pattern-level to structural-level thinking, but sometimes the best action we can take must remain focused on the present, at the event level—for example, when a building is aflame, the highest leverage action in the moment is to react by putting out the fire. Any other action would be downright inappropriate. But, if that's all we did, the actions would be considered low leverage from a long-term perspective. The art of thinking at the systemic structure level comes with knowing when to address a problem at the event, pattern, or structural level, and when to use an approach that combines the three.

Figure 1.8, "Levels of Understanding," depicts the richness of these three levels of understanding.

FIGURE 1.8

Levels of Understanding

|           | Action Mode    | Time Orientation | Way of Perceiving                                     | Questions You Would Ask                                            |
|-----------|----------------|------------------|-------------------------------------------------------|--------------------------------------------------------------------|
| Events    | React!         | Present          | Witness event                                         | "What's the fastest way to react to this event now?"               |
| Patterns  | Adapt!         |                  | Measure or track patterns of events                   | "What kinds of trends or patterns of events seem to be recurring?" |
| Structure | Create Change! | Future           | Causal loop diagrams and other systems thinking tools | "What structures are in place that are causing these patterns?"    |